

ACCOUNTING AUDIT REPORT

Automated Anomaly Detection Summary

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Dataset: transaction.csv (Sample of Accounting Transactions)

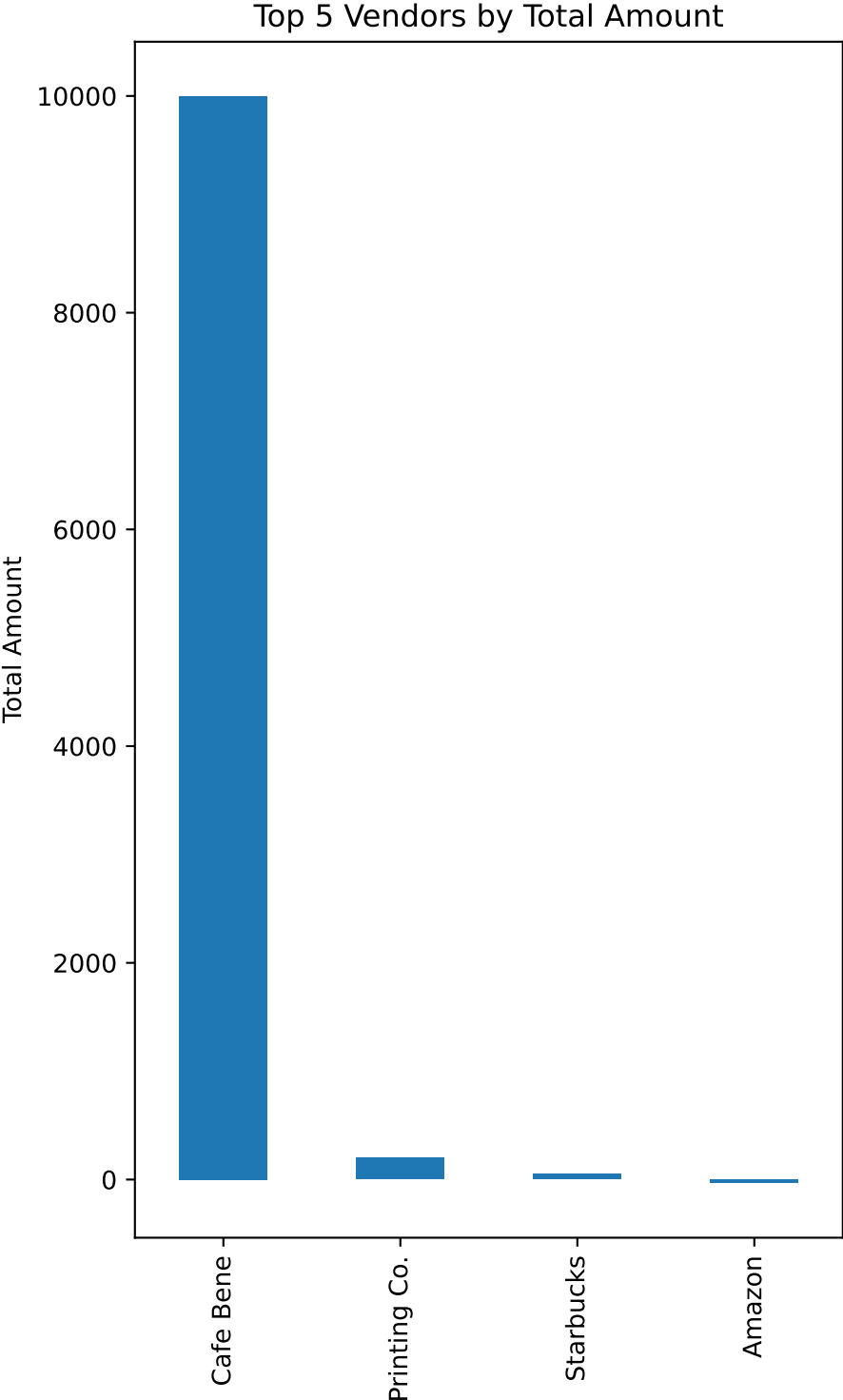
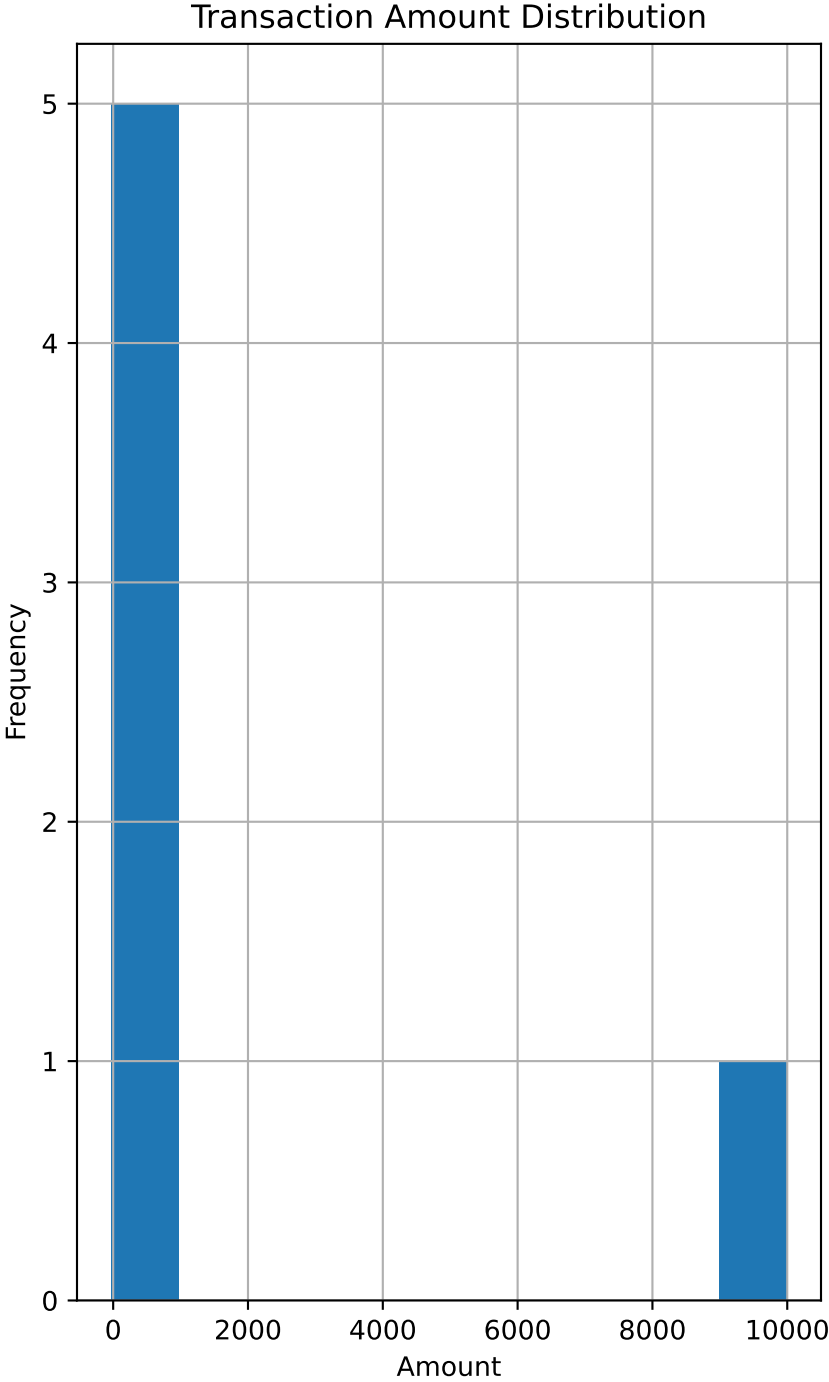
EXECUTIVE SUMMARY

This automated audit analyzes financial transaction data to detect potential anomalies such as duplicate invoices, negative amounts, missing fields, outliers, weekend transactions, and unusually high-value entries. The purpose is to assist accounting teams in quickly identifying data integrity risks and operational irregularities.

Detected Anomalies Summary:

- Duplicate Invoices: 2
- Negative Amounts: 1
- Missing Values: 1
- Outliers (z-score ≥ 3): 0
- Outliers (IQR): 1
- Over Threshold ($|Amount| > 1000$): 1
- Weekend Transactions: 2
- Empty Vendor: 1

Visual Insights



Top 3 Anomalies

These represent transactions requiring immediate review or confirmation.

Invoice_ID: INV001
Vendor: Starbucks | Amount: 25.5
Category: Marketing | Date: 2025-09-01

Invoice_ID: INV001
Vendor: Starbucks | Amount: 25.5
Category: Marketing | Date: 2025-09-08

Invoice_ID: INV002
Vendor: Amazon | Amount: -35.0
Category: Supplies | Date: 2025-09-01

Recommendations

- 1) Enforce mandatory Vendor field validation.
- 2) Check for duplicate invoice IDs before posting transactions.
- 3) Review and approve all weekend or high-value transactions.
- 4) Investigate missing fields for data entry issues.
- 5) Implement automated alerts for statistical outliers.

Appendix: Anomaly Tables (First 10 Rows Each)

Duplicate Invoices

```
{'Invoice_ID': 'INV001', 'Date': '2025-09-01', 'Vendor': 'Starbucks', 'Amount': 25.5, 'Category': 'Marketing'}  
{'Invoice_ID': 'INV001', 'Date': '2025-09-08', 'Vendor': 'Starbucks', 'Amount': 25.5, 'Category': 'Marketing'}
```

Negative Amounts

```
{'Invoice_ID': 'INV002', 'Date': '2025-09-01', 'Vendor': 'Amazon', 'Amount': -35, 'Category': 'Supplies'}
```

Missing Values

```
{'Invoice_ID': 'INV005', 'Date': '2025-09-06', 'Vendor': nan, 'Amount': 150, 'Category': 'Equipment'}
```

Outliers (IQR)

```
{'Invoice_ID': 'INV006', 'Date': '2025-09-07', 'Vendor': 'Cafe Bene', 'Amount': 9999, 'Category': 'Food'}
```

Weekend Transactions

```
{'Invoice_ID': 'INV005', 'Date': Timestamp('2025-09-06 00:00:00'), 'Vendor': nan, 'Amount': 150, 'Category': 'Equipment'}  
{'Invoice_ID': 'INV006', 'Date': Timestamp('2025-09-07 00:00:00'), 'Vendor': 'Cafe Bene', 'Amount': 9999, 'Category': 'Food'}
```